

Genetic Evaluation of a Multigenerational Neurological Disorder in a Non-Consanguineous Family

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1. Background

This report summarizes the family history, genetic testing findings, and candidate gene analysis for a 23-year-old boy with intellectual disability, developmental delay, and seizures. The purpose of this report is to explain what genes were found, how well they match the family's symptoms, and what questions remain before a final diagnosis can be confirmed.

The studied family presents with a multigenerational neurological disorder characterized predominantly by developmental delay, intellectual disability, and seizures. According to the available clinical history and pedigree, the disorder affects multiple male individuals, including the 23-year-old proband, at the time of the genetic evaluation and his affected younger brother, who is 20 years old, as well as their affected maternal uncle in the previous generation. The proband and his affected younger brother have one additional brother and two younger twin sisters, all of whom died. Several other female relatives are clinically unaffected. The family is described as non-consanguineous (Figure 1).

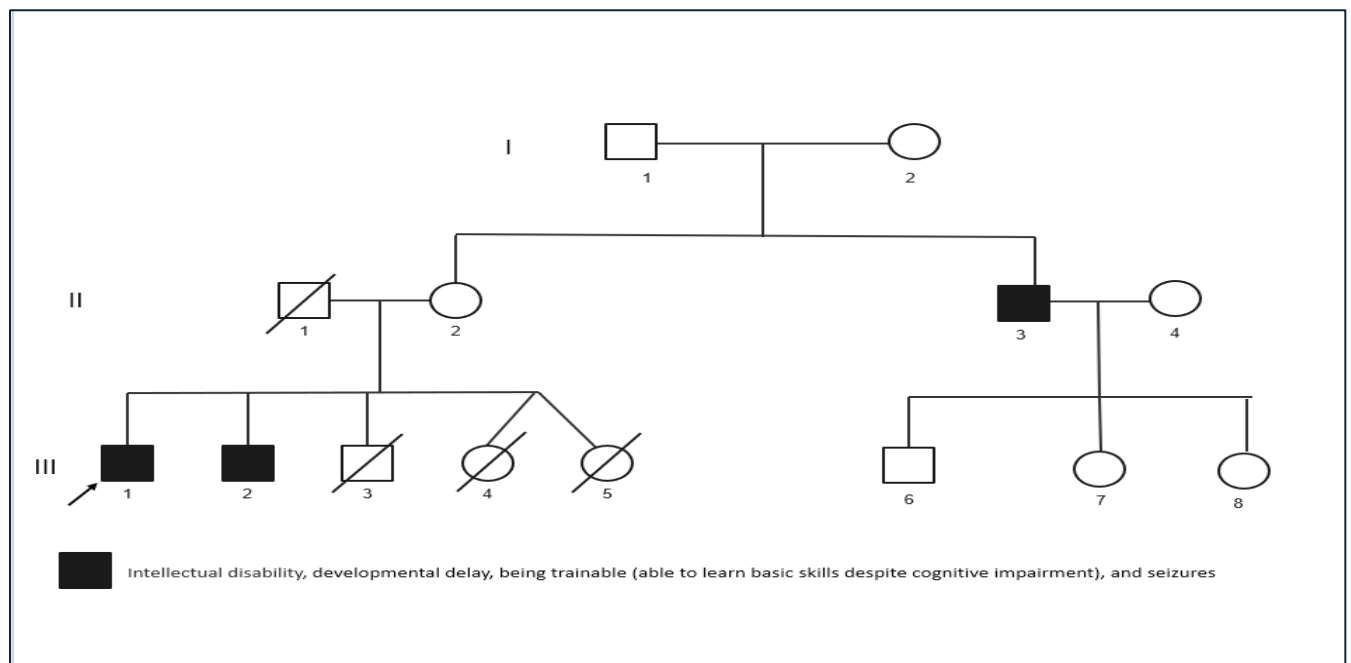


Figure 1. Three-generation pedigree showing two affected brothers (III-1, proband, indicated by arrow; III-2) and their affected maternal uncle (II-3). Filled symbols = affected; slashed symbols = deceased.

The observed phenotype is compatible with a broad group of inherited neurodevelopmental and neurological disorders. Developmental delay and intellectual disability accompanied by epilepsy can occur in a wide range of genetic conditions involving neuronal development, intracellular trafficking, chromatin regulation, synaptic function, and other cellular pathways. Therefore, both the clinical phenotype and the pattern of inheritance observed in the pedigree were considered during the evaluation of candidate genes and variants.

2. Methodology

Targeted gene panel sequencing was performed on the proband. The resulting variant data were analyzed in Microsoft Excel. First, the Excel file was filtered to exclude benign and likely benign variants, synonymous mutations, and other low-priority variants. After this filtering step, a set of candidate genes remained. These candidates were further evaluated and prioritized using publicly available databases and resources, including OMIM, ClinVar, the ACMG/AMP guidelines, NCBI, PubMed, Ensembl, and gnomAD. Through this process, two genes of particular interest were identified: *AP4M1* and *KMT2D*.

3. Candidate Genes and Variants

3.1 AP4M1 and the variant c.1342G>A (p.Ala448Thr)

The *AP4M1* gene (OMIM 602296) [1] encodes the mu-1 subunit of the adaptor protein complex 4 (AP-4). AP-4 is a heterotetrameric complex that mediates vesicle trafficking from the trans-Golgi network to the endosomal-lysosomal system. This trafficking pathway is essential for proper neuronal development and function. Biallelic pathogenic variants in any of the four AP-4 subunit genes (*AP4B1*, *AP4E1*, *AP4M1*, or *AP4S1*) cause AP-4-associated hereditary spastic paraplegia (AP-4-HSP) [2].

Pathogenic variants in *AP4M1* specifically cause spastic paraplegia 50 (SPG50; OMIM 612936) [3]. The phenotype is characterized by neonatal or infantile hypotonia that progresses to hypertonia and spasticity, severe intellectual disability with poor or absent speech development, microcephaly, seizures, and variable brain imaging findings such as white matter abnormalities, thin corpus callosum, ventriculomegaly, and cerebellar hypoplasia or atrophy [2,4,5]. Developmental delay, intellectual disability, and seizures are core and frequently reported features [4,5]. The disorder is inherited in an autosomal recessive manner [2].

The variant identified in this family is NM_004722.4:c.1342G>A (p.Ala448Thr) [6,7]. This missense change substitutes alanine with threonine at amino acid position 448 of the AP-4 complex subunit mu-1 protein. The variant is catalogued in ClinVar under Variation ID 210207 [6]. Aggregate classifications in ClinVar report conflicting interpretations of pathogenicity, ranging from uncertain significance to likely benign. Population frequency data indicate that the allele is rare (gnomAD allele frequency approximately 0.00023).

An additional heterozygous AP4M1 change was identified at chr7:99,703,958 (T>C), located in a non-coding (intronic) region of the gene. This variant is common in the general population (gnomAD allele frequency $\approx$ 30%), is not predicted to alter splicing, and has no ClinVar disease classification on record. It therefore does not constitute a plausible second (“in trans”) pathogenic allele and does not change the overall interpretation that biallelic pathogenicity of AP4M1 remains unconfirmed in this proband.

Given the autosomal recessive nature of SPG50, a single heterozygous occurrence of this variant would not be expected to fully explain the pedigree pattern observed in this family; however, it remains a candidate for further segregation analysis, especially if compound heterozygosity or additional variants in the same pathway are identified.

All in all, AP4M1 is a strong phenotype-and inheritance compatible candidate gene however the identified variant has conflicting likely benign classifications and therefore does not currently provide sufficient evidence for causality.

3.2 KMT2D and the variant c.8924G>A

The *KMT2D* gene (weaker candidate for this family)(OMIM 602113) [8], also known as MLL2 or MLL4, encodes lysine-specific methyltransferase 2D. This enzyme is a major histone H3 lysine 4 (H3K4) mono-methyltransferase and functions as a transcriptional regulator during embryonic development. Heterozygous pathogenic variants in *KMT2D* are the primary molecular cause of Kabuki syndrome type 1 (KS1; OMIM 147920) [9], an autosomal dominant multiple congenital anomaly syndrome.

Kabuki syndrome is characterized by distinctive facial features, mild-to-moderate intellectual disability, postnatal growth deficiency, skeletal anomalies, persistence of fetal fingertip pads, and variable congenital malformations. Neurologic features frequently include developmental delay, intellectual disability, hypotonia, and seizures [9,10]. The clinical presentation is highly variable; some individuals exhibit primarily neurologic and developmental features with milder or less classic facial dysmorphism [9].

The variant identified in this family is NM_003482:c.8924G>A, recorded in ClinVar under Variation ID 196893 [11]. Because Kabuki syndrome follows an autosomal dominant inheritance pattern with high penetrance but variable expressivity, a heterozygous *KMT2D* variant is a biologically plausible candidate to explain neurological and developmental findings transmitted across generations in a non-consanguineous pedigree. Detailed clinical phenotyping (facial features, growth parameters, congenital anomalies, and developmental history) together with segregation analysis are required to assess the contribution of this variant.

3.1 Comparison of AP4M1 and KMT2D

Table 1 summarizes the key differences and similarities between the two candidate genes and the specific variants identified in this study.

Feature	<i>AP4M1</i>	<i>KMT2D</i>
Gene / Protein function	Encodes mu-1 subunit of AP-4 complex; vesicle trafficking (TGN → endosome/lysosome); neuronal development [1,2]	Encodes histone H3K4 methyltransferase; transcriptional regulation during embryonic development [8]
Associated disease	Spastic paraplegia 50 (SPG50); AP-4-HSP [2,3]	Kabuki syndrome type 1 (KS1) [9,10]
Inheritance	Autosomal recessive [2]	Autosomal dominant [9]
Core phenotype	Developmental delay, intellectual disability, seizures, progressive spasticity, microcephaly [2,4,5]	Developmental delay, intellectual disability, hypotonia, seizures (subset), distinctive facial features, growth deficiency [9,10]
Overlap with family phenotype	Good for DD + ID + seizures; progressive spasticity not reported in family	Good for DD + ID + seizures; facial features / growth issues not reported in family
Identified variant	NM_004722.4:c.1342G>A (p.Ala448Thr) [6,7]	NM_003482:c.8924G>A [11]
ClinVar ID / Classification	Variation ID 210207; Conflicting (VUS / Likely benign) [6]	Variation ID 196893 [11]
Population frequency	Rare (gnomAD ~0.00023)	Not detailed in current data
Fit with pedigree pattern	Poor (AR; usually requires biallelic variants)	Moderate–Poor (AD; usually affects both sexes)

Table 1. Comparison of *AP4M1* and *KMT2D* genes and the variants identified in this study.

4. Discussion and Next Steps

According to the pedigree, the inheritance pattern initially appeared consistent with an X-linked disorder. However, systematic review of the variant data in the Excel file did not identify any candidate X-linked genes or variants that segregated appropriately with the neurological phenotype. Subsequent prioritization of autosomal candidates highlighted rare variants in *AP4M1* (associated with recessive SPG50) [2–5] and *KMT2D* (associated with dominant Kabuki syndrome) [9,10]. Both genes are associated with developmental delay, intellectual disability, and seizures, features that overlap with the phenotype observed in this family [4,5,9,10]. However, the established inheritance patterns of these genes differ from the male-predominant pattern suggested by the pedigree.

Further work should include: (1) detailed clinical re-evaluation of affected and unaffected family members for features of SPG50 or Kabuki syndrome; (2) segregation analysis of the identified variants; (3) assessment of zygosity and potential compound heterozygosity for the *AP4M1* allele; and (4) functional or bioinformatic prediction studies to refine the pathogenicity classification of both variants according to the ACMG/AMP guidelines [12].

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